LAPORAN JOBSHEET
“7-API Routes pada Next.js dan Integrasi Firebase”
MATA KULIAH PEMROGRAMAN BERBASIS FRAMEWORK

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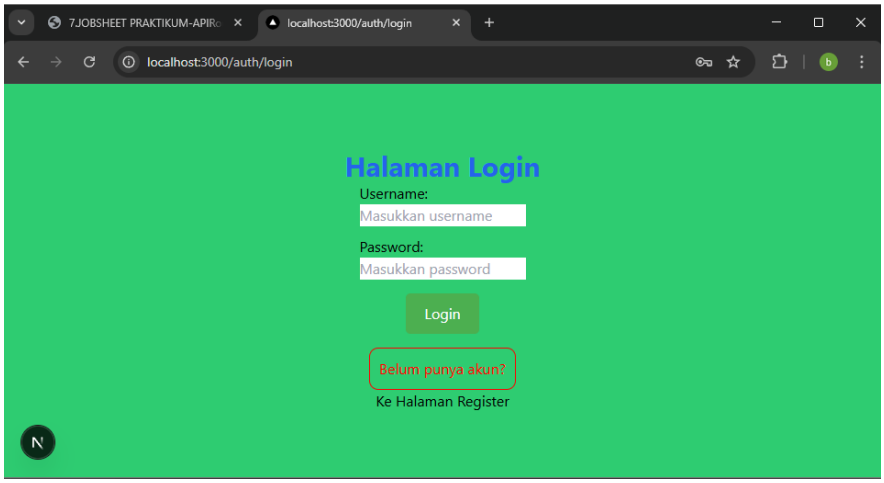
PROGRAM STUDI D-IV TEKNIK INFORMATIKA

JURUSAN TEKNOLOGI INFORMASI

POLITEKNIK NEGERI MALANG

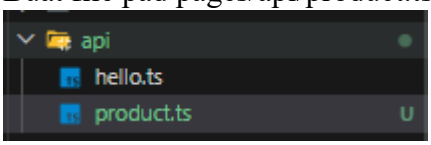
2026

Langkah 1 – Menjalankan Project



Langkah 2 – Membuat API Produk

1. Buat file pad pages/api/product.ts

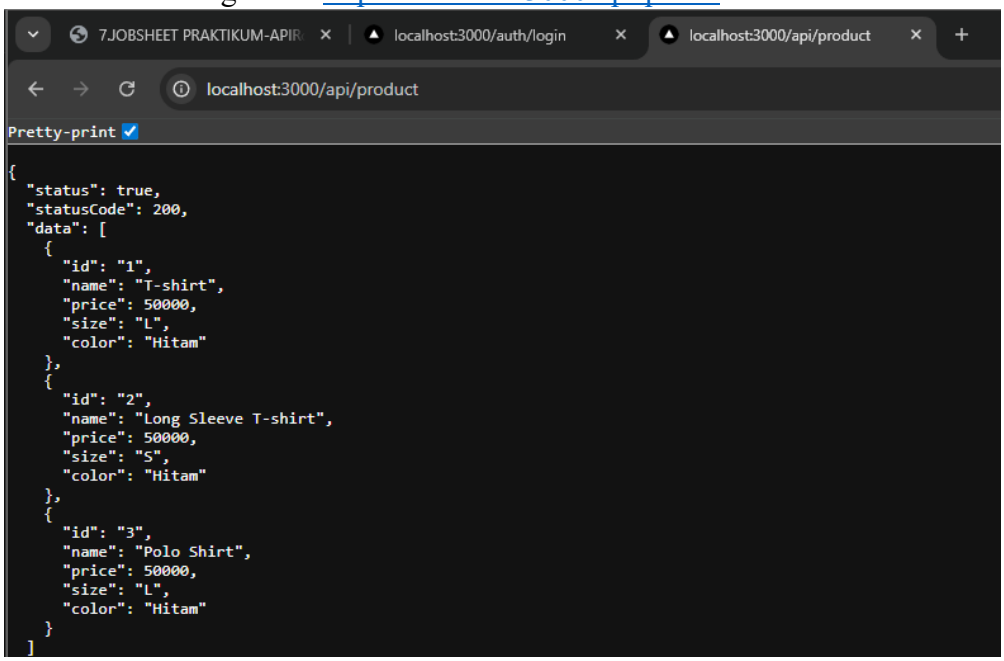


2. Modifikasi kode di dalam product.ts

P2 > Praktikum > next-routing > src > pages > api > product.ts > handler > data

```
import type { NextApiRequest, NextApiResponse } from "next";

type Data = {
  status: boolean;
  statusCode: number;

  data: {
    id: string;
    name: string;
    price: number;
    size: string;
    color: string;
  }[];
};

export default function handler(
  req: NextApiRequest,
  res: NextApiResponse<Data>,
) {
  const data = [
    {
      id: "1",
      name: "T-shirt",
      price: 50000,
      size: "L",
      color: "Hitam",
    },
    {
      id: "2",
      name: "Long Sleeve T-shirt",
      price: 50000,
      size: "S",
      color: "Hitam",
    },
    {
      id: "3",
      name: "Polo Shirt",
      price: 50000,
      size: "L",
      color: "Hitam",
    },
  ];
  res.status(200).json({ status: true, statusCode: 200, data });
}
```

3. Hasil ketika mengakses: http://localhost:3000/api/product




```
{
  "status": true,
  "statusCode": 200,
  "data": [
    {
      "id": "1",
      "name": "T-shirt",
      "price": 50000,
      "size": "L",
      "color": "Hitam"
    },
    {
      "id": "2",
      "name": "Long Sleeve T-shirt",
      "price": 50000,
      "size": "S",
      "color": "Hitam"
    },
    {
      "id": "3",
      "name": "Polo Shirt",
      "price": 50000,
      "size": "L",
      "color": "Hitam"
    }
  ]
```


Langkah 3 – Fetch Data API di Frontend

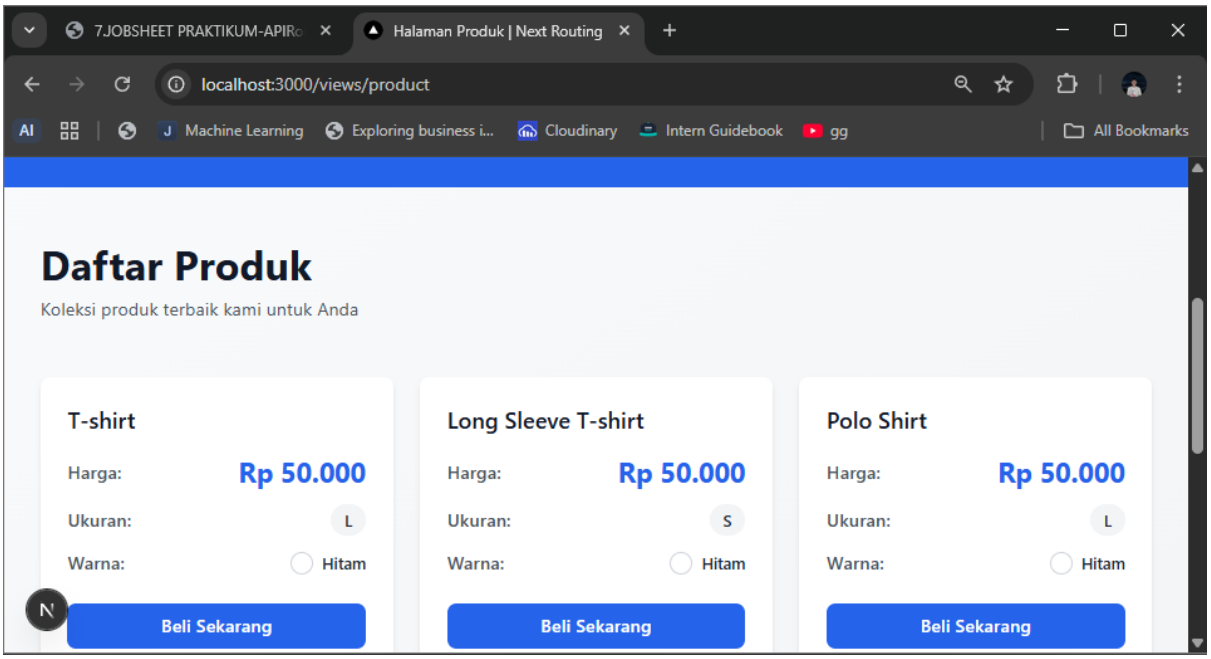
1. Modifikasi index.tsx di file product

P2 > Praktikum > next-routing > src > pages > views > product > components > MiddleSection.tsx > useCategory > useEffect() callback

```
import { useState, useEffect } from "react";

type ProductType = {
  id: string;
  name: string;
  price: number;
  size: string;
  color: string;
};
const useCategory = () => {
  const [products, setProducts] = useState([]);

  useEffect(() => {
    const fetchProducts = async () => {
      try {
        const response = await fetch("/api/product");
        const data = await response.json();
        setProducts(data.data);
      } catch (error) {
        console.error("Error fetching products:", error);
      }
    };

    fetchProducts();
  }, []);

  return {
    products,
  };
};
```

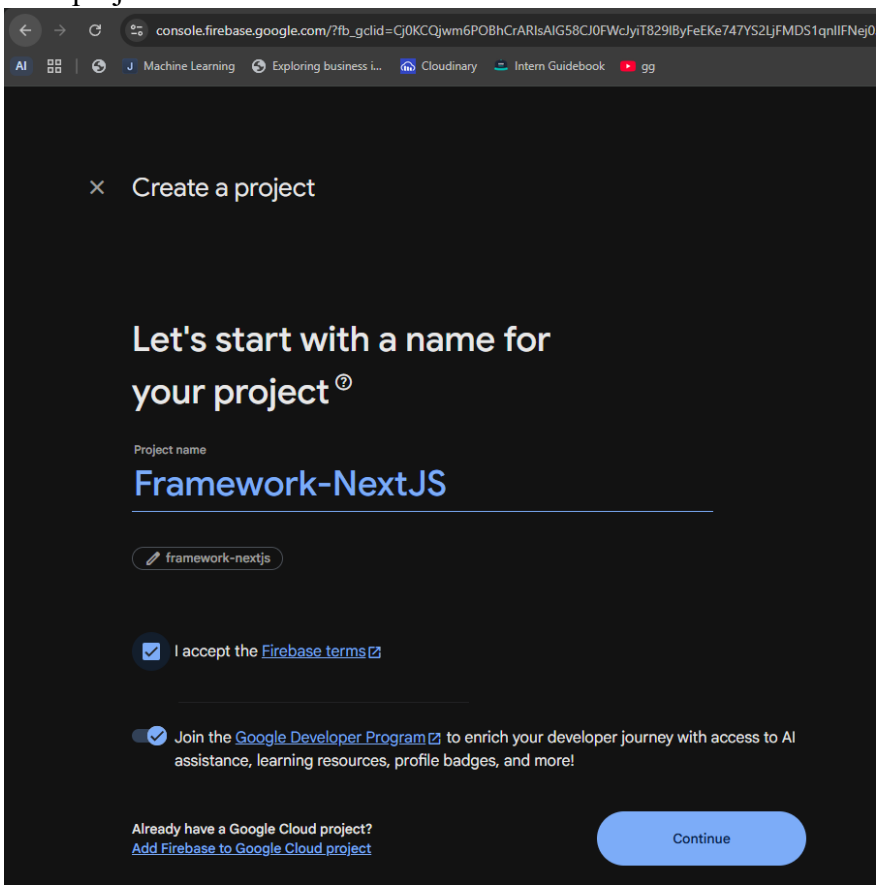
Karena disini saya menggunakan beberapa section Fetch Data API saya taruh di MiddleSection, karena sudah ada Greetings pada HeroSection.

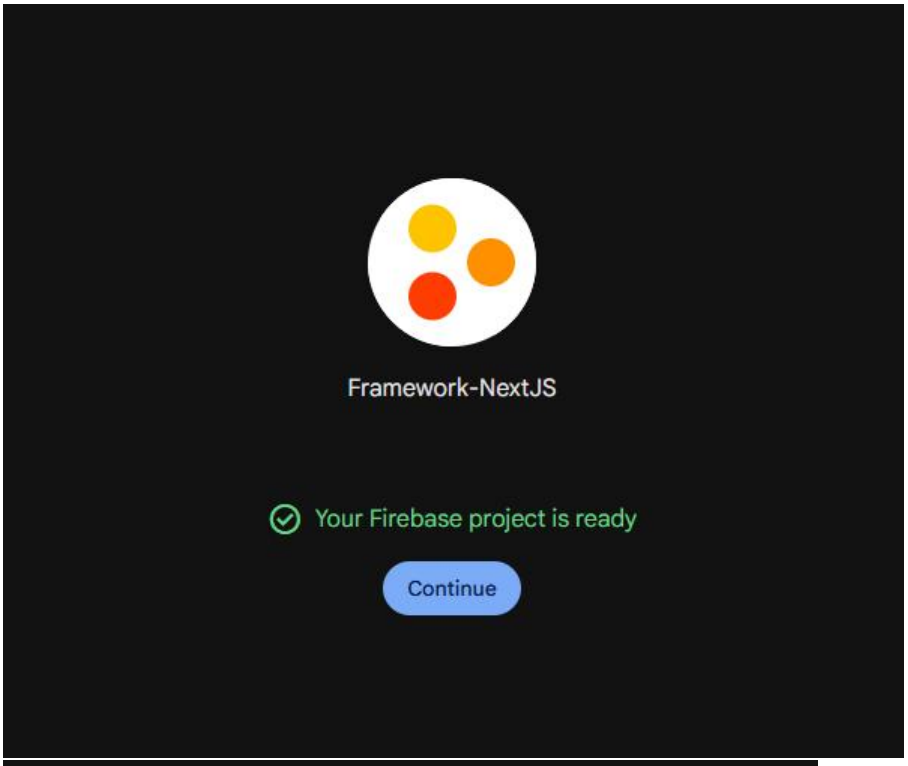
- Ketika Cek di Browser

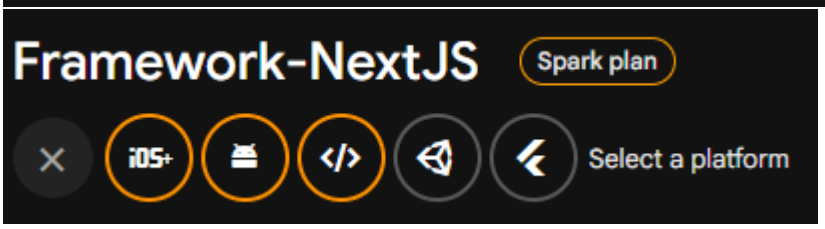


Langkah 4 - Setup Firebase

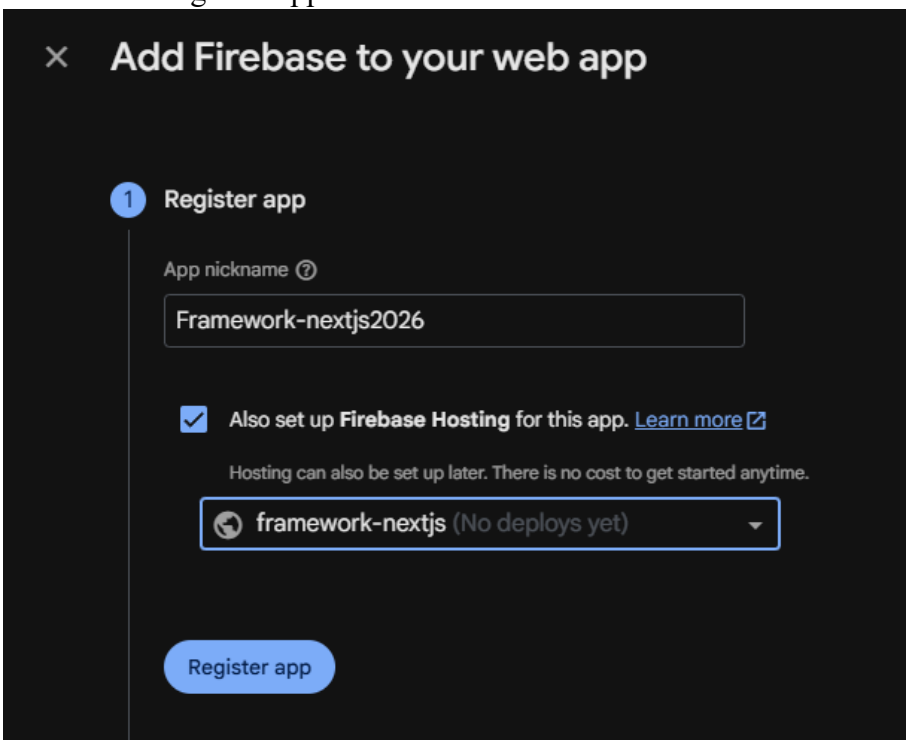
1. Buka Firebase Go To Console (login dengan login google)
2. Buat project baru



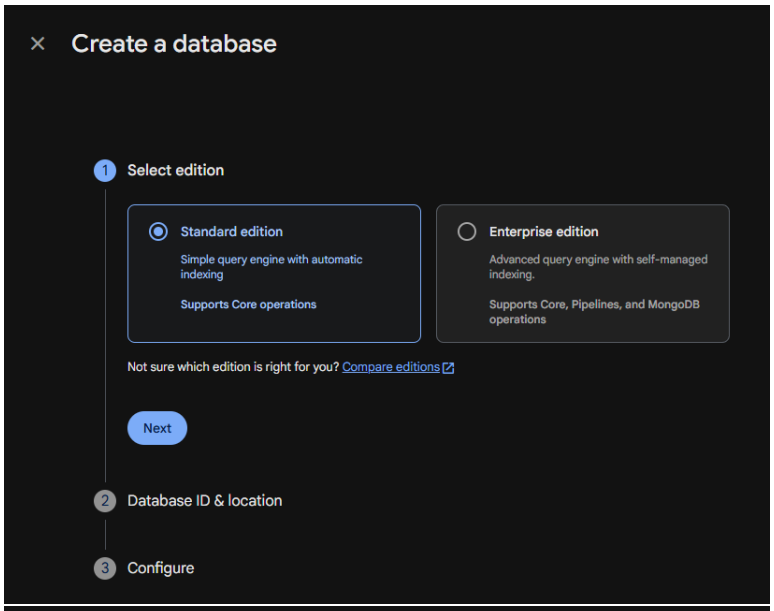


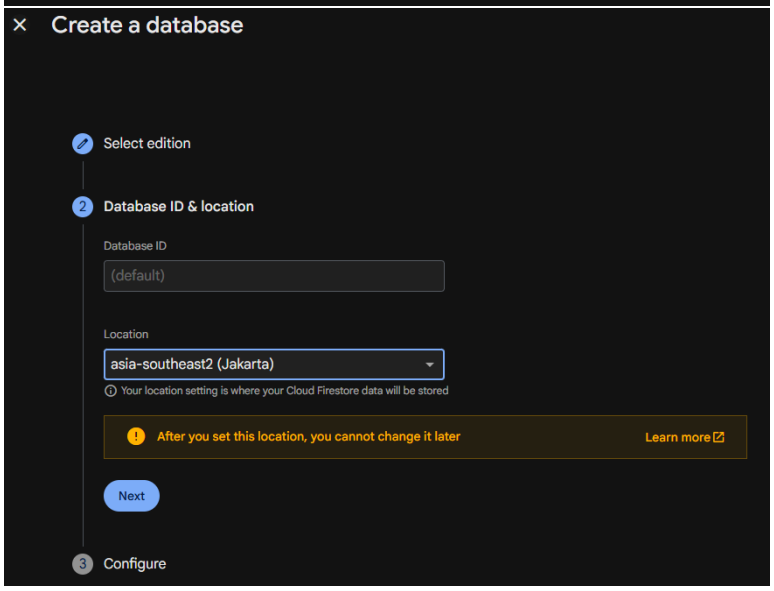


Melakukan Register App untuk web

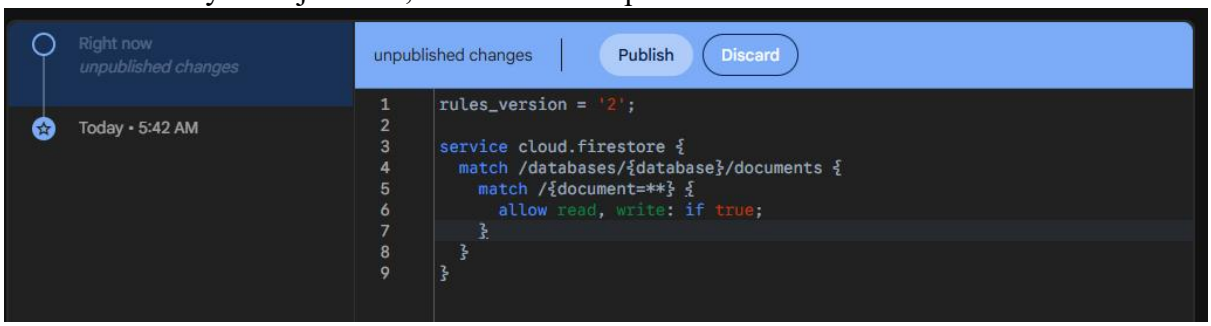


3. Membuat Database



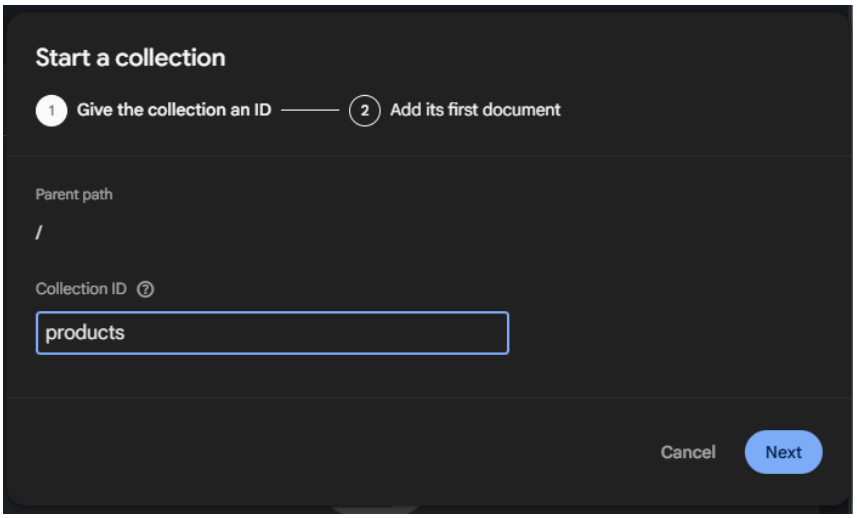


Merubah rulesnya menjadi true, kemudian click publish

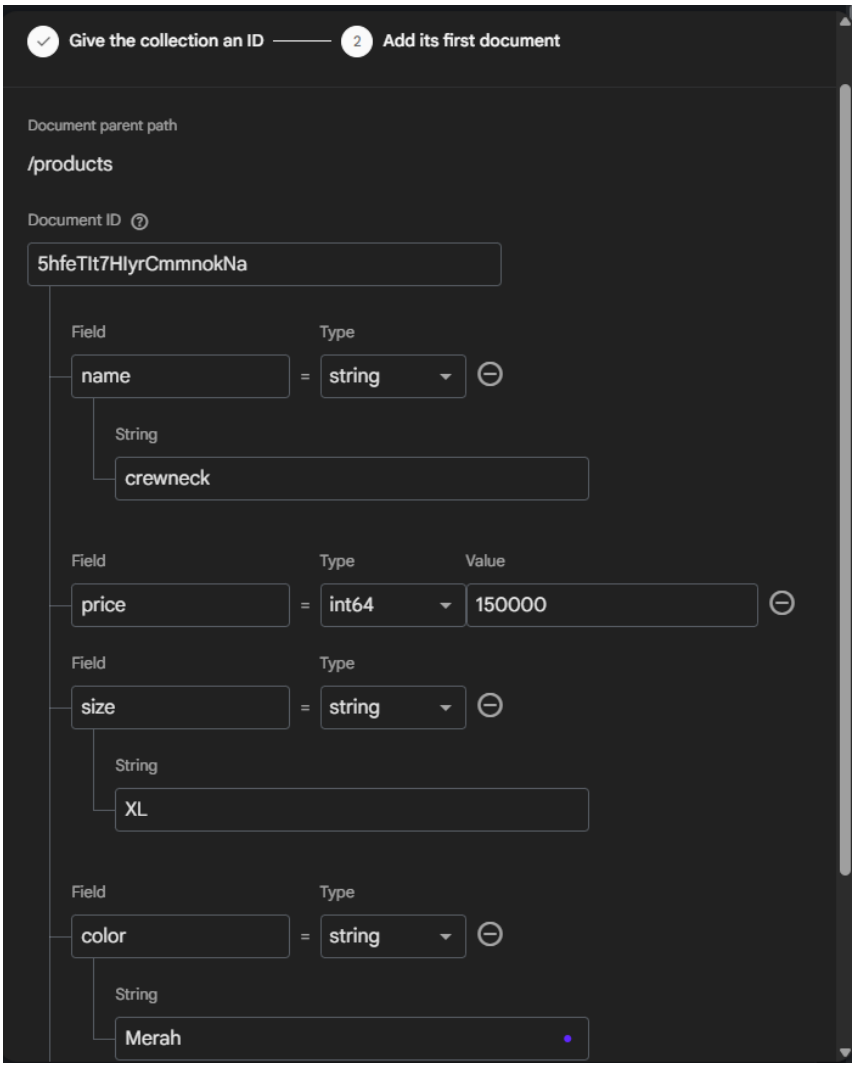


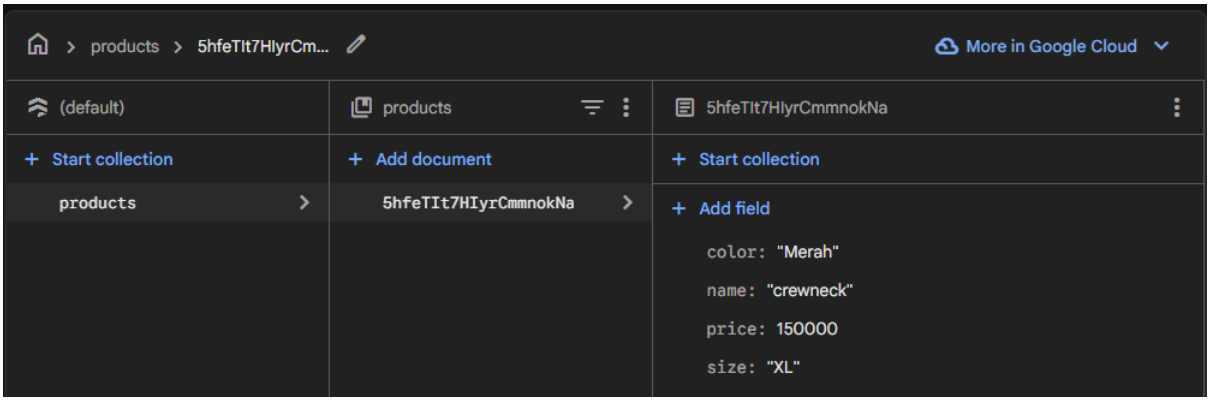
4. Buat Collection:

5. Products



6. Menambahkan fields



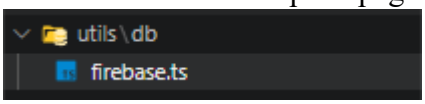


Langkah 5 – Install Firebase

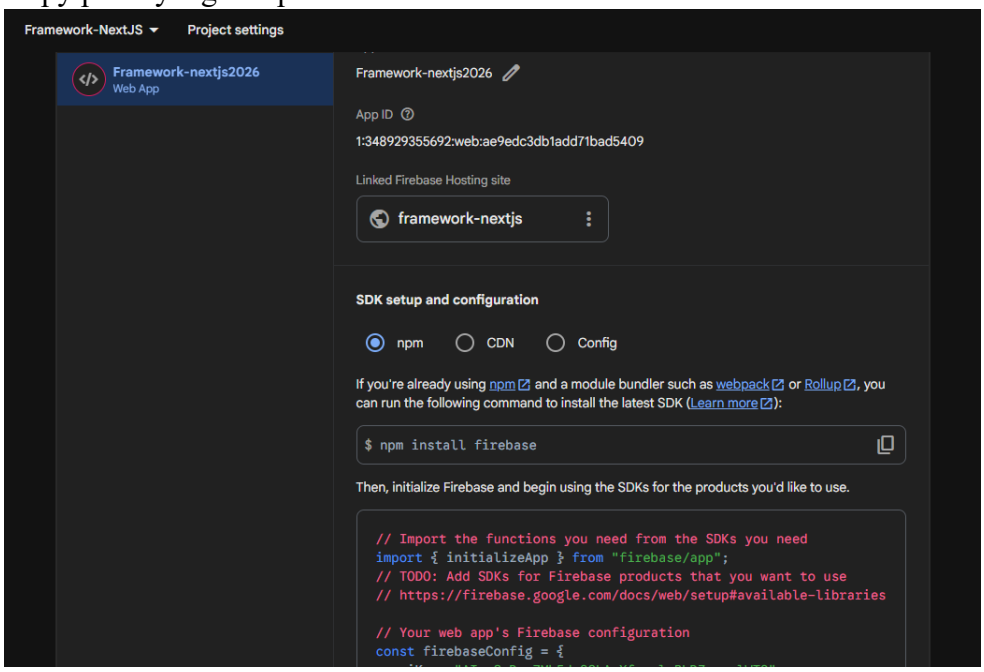
1. Npm install firebase

```
PS D:\KULIAH\SEMESTER-6\Pemrograman-Berbasis-Framework\P2\Praktikum\next-routing> npm install firebase

added 79 packages, and audited 470 packages in 37s

154 packages are looking for funding
  run `npm fund` for details

4 vulnerabilities (2 moderate, 2 high)

To address issues that do not require attention, run:
  npm audit fix

To address all issues, run:
  npm audit fix --force

Run `npm audit` for details.
PS D:\KULIAH\SEMESTER-6\Pemrograman-Berbasis-Framework\P2\Praktikum\next-routing> npm audit fix

changed 6 packages, and audited 470 packages in 5s

154 packages are looking for funding
  run `npm fund` for details

# npm audit report

next  16.0.0-beta.0 - 16.1.6
Severity: moderate
```

2. Buat folder dan file ts pada pages utlis/db/firebase.ts

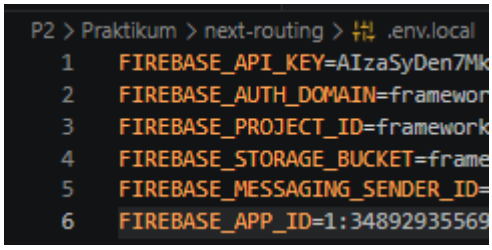


3. Copy paste yang ada pada kotak merah ke file firebase.ts



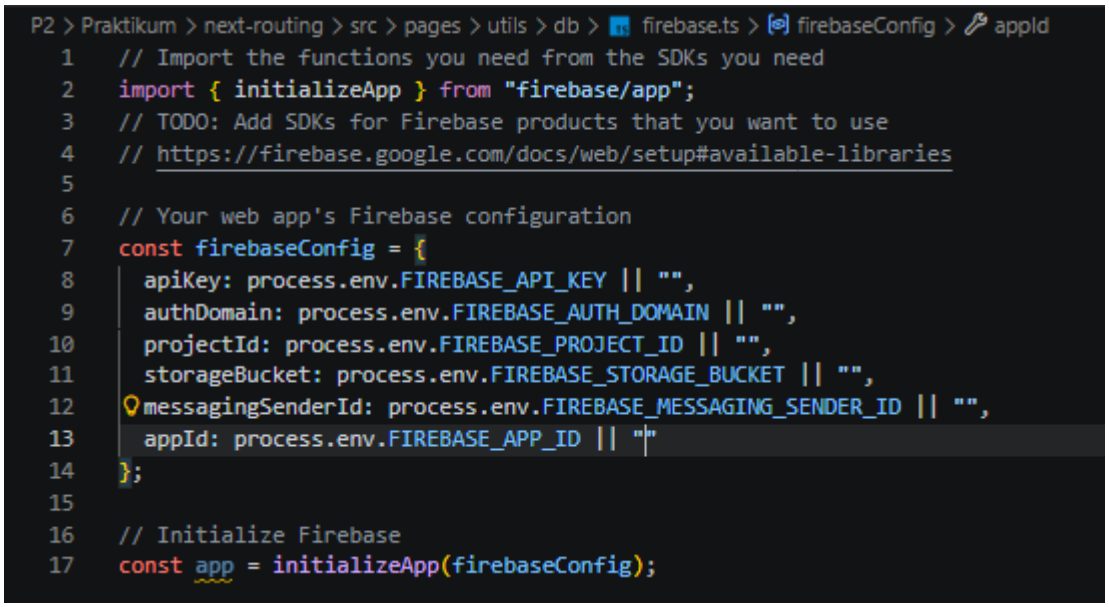
Langkah 6 - Konfigurasi Environment Variable agar credensial firebase tidak dapat dilihat saat dipush di repository

1. Buat file: .env.local



Langkah 7 – Konfigurasi Firebase

1. Modifikasi firebase.ts



Langkah 8 - Ambil Data dari Firestore

1. Buat file utils/db/servicefirebase.ts

2. Modifikasi servicesfirebase.ts

```
P2 > Praktikum > next-routing > src > pages > utils > db > servicesfirebase.ts > ...
import { getFirestore, collection, getDocs } from "firebase/firestore";
import app from "./firebase";

const db = getFirestore(app);

export async function retrieveProducts(collectionName: string) {
  const snapshot = await getDocs(collection(db, collectionName));
  const data = snapshot.docs.map((doc) => ({ id: doc.id, ...doc.data() }));
  return data;
}
```

Langkah 9 - API Mengambil Data Firebase

1. Edit pages/api/product.ts

```
Praktikum > next-routing > src > pages > api > product.ts > Data
import type { NextApiRequest, NextApiResponse } from "next";
import { retrieveProducts } from "../utils/db/servicesfirebase";


type Data = {
  status: boolean;
  statusCode: number;
  /* eslint-disable @typescript-eslint/no-explicit-any */
  data: any;
};

export default async function handler(
  req: NextApiRequest,
  res: NextApiResponse<Data>,
) {
  const data = await retrieveProducts("products");
  res.status(200).json({ status: true, statusCode: 200, data });
}
```

2. Cek di browser untuk cek API nya

localhost:3000/api/product

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```
{
    "status": true,
    "statusCode": 200,
    "data": [
        {
            "id": "5hfeTIt7HIyrCmmnokNa",
            "color": "Merah",
            "size": "XL",
            "name": "crewneck",
            "price": 150000
        }
    ]
}
```


3. Modifikasi index.ts pada produk sesuaikan nama typenya dan db nya

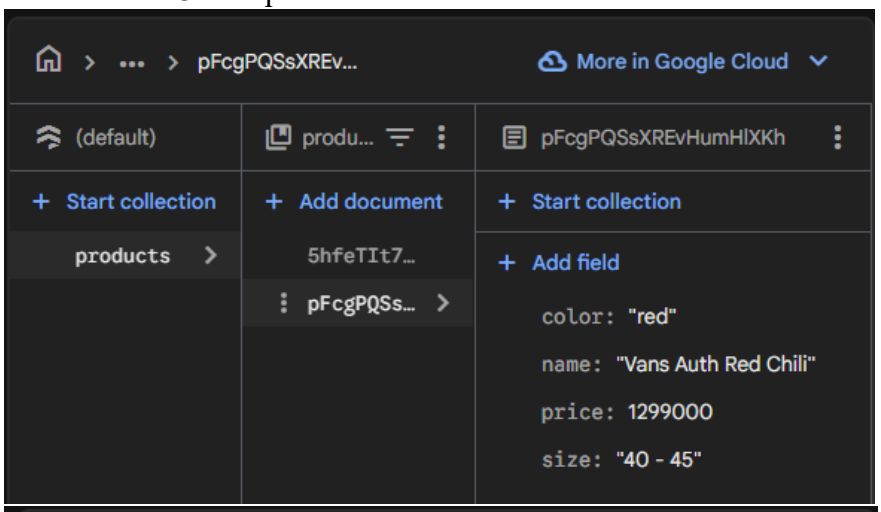
```
import { useState, useEffect } from "react";

type ProductType = {
  id: string;
  name: string;
  price: number;
  size: string;
  color: string;
};
```

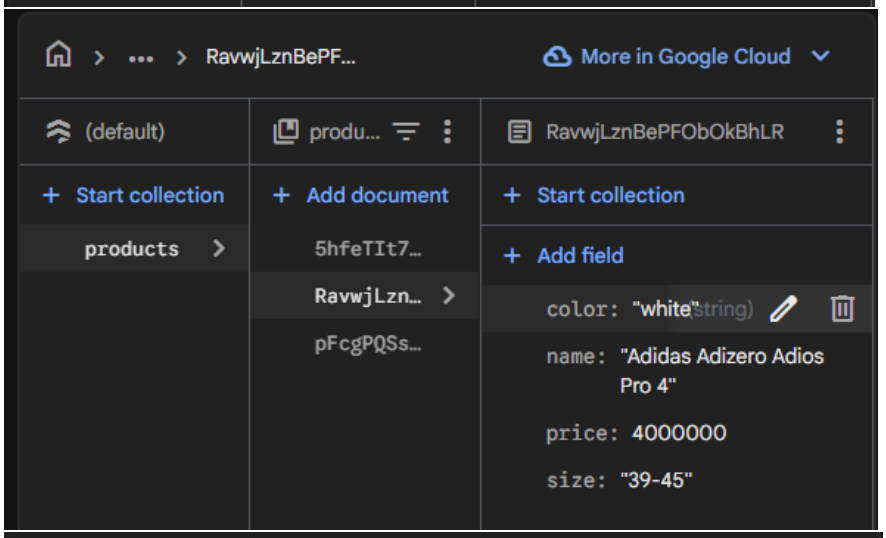
cek di browser

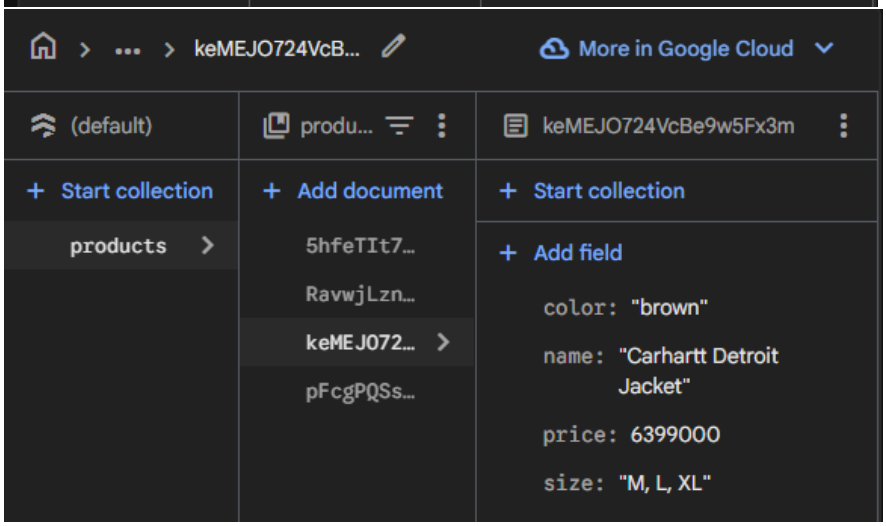
```
<div className="p-6">
  <h2 className="text-xl font-semibold text-gray-900 mb-4">
    {product.name}
  </h2>
  <div className="space-y-3">
    <div className="flex justify-between items-center">
      <span className="text-gray-600 font-medium">Harga:</span>
      <span className="text-2xl font-bold text-blue-600">
        Rp {product.price.toLocaleString("id-ID")}
      </span>
    </div>
    <div className="flex justify-between items-center">
      <span className="text-gray-600 font-medium">Ukuran:</span>
      <span className="px-3 py-1 bg-gray-100 text-gray-800 rounded-full tex
        {product.size}
      </span>
    </div>
    <div className="flex justify-between items-center">
      <span className="text-gray-600 font-medium">Warna:</span>
      <div className="flex items-center gap-2">
        <div
          className="w-5 h-5 rounded-full border-2 border-gray-300"
          style={{
            backgroundColor: product.color.toLowerCase(),
          }}
        ></div>
        <span className="text-gray-800 text-sm font-medium">
          {product.color}
        </span>
      </div>
    </div>
  </div>
```

TUGAS PRAKTIKUM

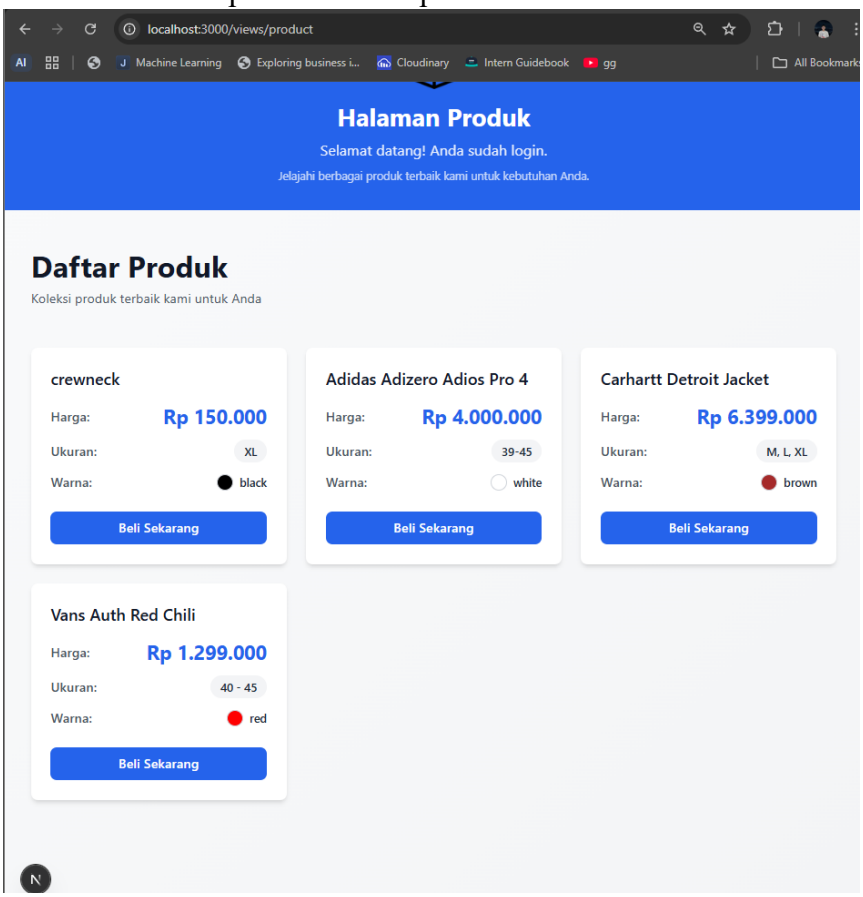
1. Tugas 1 (wajib)
   - Tambahkan minimal 3 data produk di Firestore







- Pastikan data tampil di halaman produk

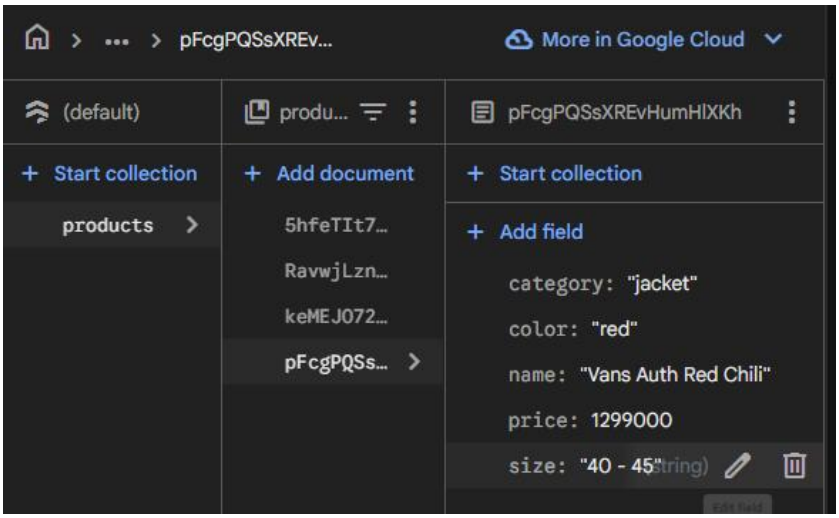


2. Tugas 2 (Wajib)
   - Tambahkan field category

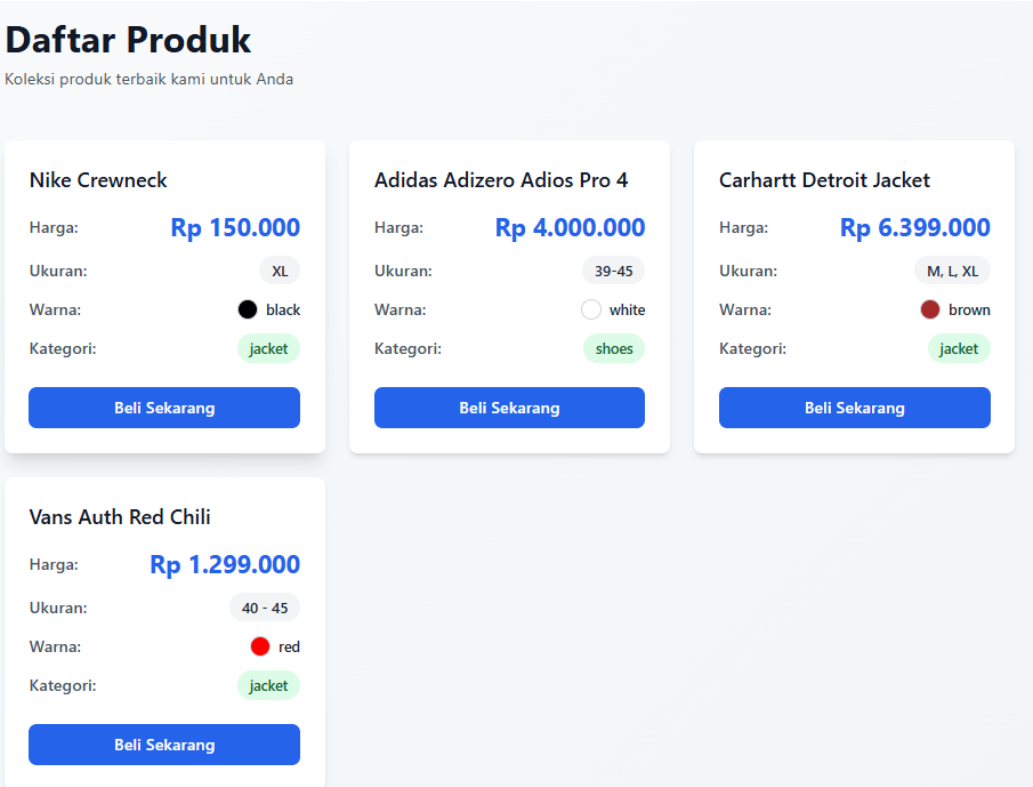
```
type ProductType = {
  id: string;
  name: string;
  price: number;
  size: string;
  color: string;
  category: string;
};
```

```
<div className="flex justify-between items-center">
  <span className="text-gray-600 font-medium">
    Kategori:
  </span>
  <span className="px-3 py-1 bg-green-100 text-green-800 rounded-full text-
    {product.category}
  </span>
</div>
```

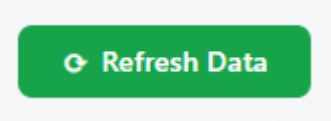
Di firebase



- Tampilkan category di frontend



3. Tugas 3 (Pengayaan)
   - Tambahkan tombol Refresh Data



- Gunakan fetch ulang tanpa reload halaman

```
const fetchProducts = async () => {
  setLoading(true);
  try {
    const response = await fetch("/api/product");
    const data = await response.json();
    setProducts(data.data);
  } catch (error) {
    console.error("Error fetching products:", error);
  } finally {
    setLoading(false);
  }
};

useEffect(() => {
  fetchProducts();
}, []);
```

PERTANYAAN EVALUASI

1. Apa fungsi API Routes pada Next.js?
   API Routes berfungsi untuk membuat backend endpoint langsung di dalam project Next.js. Jadi kamu bisa handle request seperti GET, POST, dll tanpa perlu server terpisah (misalnya Express). Biasanya dipakai untuk ambil data, autentikasi, atau koneksi ke database.

2. Mengapa .env.local tidak boleh di-push ke repository?
   Karena file ini berisi data sensitif seperti API key, password, dan konfigurasi rahasia. Kalau di-push, orang lain bisa akses dan menyalahgunakan (misalnya akses database atau Firebase kamu).

3. Apa perbedaan data statis dan data dinamis?
   - Data statis: data yang tetap dan jarang berubah (contoh: konten halaman About).
   - Data dinamis: data yang berubah-ubah dan biasanya diambil dari database atau API (contoh: produk dari Firebase, data user).

4. Mengapa Next.js disebut framework fullstack?
   Karena Next.js bisa handle:
   - Frontend → tampilan UI (React)
   - Backend → API Routes, server logic
     Jadi dalam satu project kamu bisa bikin client + server sekaligus tanpa perlu framework tambahan.